

Received Order Analysis Engine (ROAE)

Analysis of the King Wen sequence including observations by Terence McKenna

Hexagram reference table

Hexagram reference table
Each hexagram is a stack of 6 lines, either solid (1/yang) or broken (0/yin).
The 'binary' column encodes these 6 lines as bits (bottom line = rightmost bit).
Each hexagram is also composed of two 3-line trigrams: an upper and a lower.
There are 8 possible trigrams (Heaven, Earth, Thunder, Water, Mountain, Wind, Fire, Lake), giving 8x8 = 64 possible hexagrams.

Pos	Hex	Binary	Upper	Lower	Name
01	䷀	111111	Qian Heaven	Qian Heaven	The Creative
02	䷁	000000	Kun Earth	Kun Earth	The Receptive
03	䷂	010001	Kan Water	Zhen Thunder	Difficulty at the Beginning
04	䷃	100010	Gen Mountain	Kan Water	Youthful Folly
05	䷄	010111	Kan Water	Qian Heaven	Waiting
06	䷅	111010	Qian Heaven	Kan Water	Conflict
07	䷆	000010	Kun Earth	Kan Water	The Army
08	䷇	010000	Kan Water	Kun Earth	Holding Together
09	䷈	110111	Xun Wind	Qian Heaven	Small Taming
10	䷉	111011	Qian Heaven	Dui Lake	Treading
11	䷊	000111	Kun Earth	Qian Heaven	Peace
12	䷋	111000	Qian Heaven	Kun Earth	Standstill
13	䷌	111101	Qian Heaven	Li Fire	Fellowship
14	䷍	101111	Li Fire	Qian Heaven	Great Possession
15	䷎	000100	Kun Earth	Gen Mountain	Modesty
16	䷏	001000	Zhen Thunder	Kun Earth	Enthusiasm
17	䷐	011001	Dui Lake	Zhen Thunder	Following
18	䷑	100110	Gen Mountain	Xun Wind	Work on the Decayed
19	䷒	000011	Kun Earth	Dui Lake	Approach
20	䷓	110000	Xun Wind	Kun Earth	Contemplation
21	䷔	101001	Li Fire	Zhen Thunder	Biting Through
22	䷕	100101	Gen Mountain	Li Fire	Grace
23	䷖	100000	Gen Mountain	Kun Earth	Splitting Apart
24	䷗	000001	Kun Earth	Zhen Thunder	Return
25	䷘	111001	Qian Heaven	Zhen Thunder	Innocence
26	䷙	100111	Gen Mountain	Qian Heaven	Great Taming
27	䷚	100001	Gen Mountain	Zhen Thunder	Nourishment
28	䷛	011110	Dui Lake	Xun Wind	Great Preponderance
29	䷜	010010	Kan Water	Kan Water	The Abysmal
30	䷝	101101	Li Fire	Li Fire	The Clinging
31	䷞	011100	Dui Lake	Gen Mountain	Influence
32	䷟	001110	Zhen Thunder	Xun Wind	Duration
33	䷠	111100	Qian Heaven	Gen Mountain	Retreat
34	䷡	001111	Zhen Thunder	Qian Heaven	Great Power
35	䷢	101000	Li Fire	Kun Earth	Progress
36	䷣	000101	Kun Earth	Li Fire	Darkening of the Light
37	䷤	110101	Xun Wind	Li Fire	The Family
38	䷥	101011	Li Fire	Dui Lake	Opposition
39	䷦	010100	Kan Water	Gen Mountain	Obstruction
40	䷧	001010	Zhen Thunder	Kan Water	Deliverance
41	䷨	100011	Gen Mountain	Dui Lake	Decrease
42	䷩	110001	Xun Wind	Zhen Thunder	Increase
43	䷪	011111	Dui Lake	Qian Heaven	Breakthrough
44	䷫	111110	Qian Heaven	Xun Wind	Coming to Meet
45	䷬	011000	Dui Lake	Kun Earth	Gathering Together
46	䷭	000110	Kun Earth	Xun Wind	Pushing Upward
47	䷮	011010	Dui Lake	Kan Water	Oppression
48	䷯	010110	Kan Water	Xun Wind	The Well
49	䷰	011101	Dui Lake	Li Fire	Revolution
50	䷱	101110	Li Fire	Xun Wind	The Cauldron
51	䷲	001001	Zhen Thunder	Zhen Thunder	The Arousing
52	䷳	100100	Gen Mountain	Gen Mountain	Keeping Still

53				110100		Xun Wind		Gen Mountain		Development
54				001011		Zhen Thunder		Dui Lake		The Marrying Maiden
55				001101		Zhen Thunder		Li Fire		Abundance
56				101100		Li Fire		Gen Mountain		The Wanderer
57				110110		Xun Wind		Xun Wind		The Gentle
58				011011		Dui Lake		Dui Lake		The Joyous
59				110010		Xun Wind		Kan Water		Dispersion
60				010011		Kan Water		Dui Lake		Limitation
61				110011		Xun Wind		Dui Lake		Inner Truth
62				001100		Zhen Thunder		Gen Mountain		Small Preponderance
63				010101		Kan Water		Li Fire		After Completion
64				101010		Li Fire		Kan Water		Before Completion

Reverse vs. Inverse pair analysis

Reverse vs. Inverse pair analysis

The 64 hexagrams are traditionally grouped into 32 pairs (1-2, 3-4, ..., 63-64). For each pair, we test two relationships:

- Reverse: flip the hexagram upside down (read the lines top-to-bottom instead of bottom-to-top). If the second hexagram upside-down equals the first, they are a reverse pair.
- Inverse: toggle every line (solid becomes broken, broken becomes solid). If toggling all lines of one hexagram gives the other, they are inverse.

Key finding: EVERY pair in the King Wen sequence is one or the other – none are unrelated. This perfect pairing structure is unlikely to occur by chance.

01  Inverse 02 

03  Reverse 04 

05  Reverse 06 

07  Reverse 08 

09  Reverse 10 

11  Reverse 12 

13  Reverse 14 

15  Reverse 16 

17  Reverse 18 

19  Reverse 20 

21  Reverse 22 

23  Reverse 24 

25  Reverse 26 

27  Inverse 28 

29  Inverse 30 

31  Reverse 32 

33  Reverse 34 

35  Reverse 36 

37  Reverse 38 

39  Reverse 40 

41  Reverse 42 

43  Reverse 44 

45  Reverse 46 

47  Reverse 48 

49  Reverse 50 

51  Reverse 52 

53  Reverse 54 

55  Reverse 56 

57  Reverse 58 

59  Reverse 60 

61  Inverse 62 

63  Reverse 64 

Reverse count pairs: 28/32 (87.5%)

Inverse count pairs: 04/32 (12.5%)

Neither count pairs: 00/32 (0.0%)

First order of difference (the 'wave')

First order of difference (the 'wave')

For each consecutive pair of hexagrams, count how many of the 6 lines change.
This produces a sequence of values from 0 (identical) to 6 (every line flipped).
The resulting 'wave' is central to analysis of the King Wen sequence.
Key observation (McKenna, The Invisible Landscape, 1975): the value 5 never appears – no consecutive hexagrams differ by exactly 5 lines. The value 0 also never appears (no duplicates).

01 ䷀ to 02 ䷁ difference 6 *****
02 ䷁ to 03 ䷂ difference 2 **
03 ䷂ to 04 ䷃ difference 4 ****
04 ䷃ to 05 ䷄ difference 4 ****
05 ䷄ to 06 ䷅ difference 4 ****
06 ䷅ to 07 ䷆ difference 3 ***
07 ䷆ to 08 ䷇ difference 2 **
08 ䷇ to 09 ䷈ difference 4 ****
09 ䷈ to 10 ䷉ difference 2 **
10 ䷉ to 11 ䷊ difference 4 ****
11 ䷊ to 12 ䷋ difference 6 *****
12 ䷋ to 13 ䷌ difference 2 **
13 ䷌ to 14 ䷍ difference 2 **
14 ䷍ to 15 ䷎ difference 4 ****
15 ䷎ to 16 ䷏ difference 2 **
16 ䷏ to 17 ䷐ difference 2 **
17 ䷐ to 18 ䷑ difference 6 *****
18 ䷑ to 19 ䷒ difference 3 ***
19 ䷒ to 20 ䷓ difference 4 ****
20 ䷓ to 21 ䷔ difference 3 ***
21 ䷔ to 22 ䷕ difference 2 **
22 ䷕ to 23 ䷖ difference 2 **
23 ䷖ to 24 ䷗ difference 2 **
24 ䷗ to 25 ䷘ difference 3 ***
25 ䷘ to 26 ䷙ difference 4 ****
26 ䷙ to 27 ䷚ difference 2 **
27 ䷚ to 28 ䷛ difference 6 *****
28 ䷛ to 29 ䷜ difference 2 **
29 ䷜ to 30 ䷝ difference 6 *****
30 ䷝ to 31 ䷞ difference 3 ***
31 ䷞ to 32 ䷟ difference 2 **
32 ䷟ to 33 ䷠ difference 3 ***
33 ䷠ to 34 ䷡ difference 4 ****
34 ䷡ to 35 ䷢ difference 4 ****
35 ䷢ to 36 ䷣ difference 4 ****
36 ䷣ to 37 ䷤ difference 2 **
37 ䷤ to 38 ䷥ difference 4 ****
38 ䷥ to 39 ䷦ difference 6 *****
39 ䷦ to 40 ䷧ difference 4 ****
40 ䷧ to 41 ䷨ difference 3 ***
41 ䷨ to 42 ䷩ difference 2 **
42 ䷩ to 43 ䷪ difference 4 ****
43 ䷪ to 44 ䷫ difference 2 **
44 ䷫ to 45 ䷬ difference 3 ***
45 ䷬ to 46 ䷭ difference 4 ****
46 ䷭ to 47 ䷮ difference 3 ***
47 ䷮ to 48 ䷯ difference 2 **
48 ䷯ to 49 ䷰ difference 3 ***
49 ䷰ to 50 ䷱ difference 4 ****
50 ䷱ to 51 ䷲ difference 4 ****
51 ䷲ to 52 ䷳ difference 4 ****
52 ䷳ to 53 ䷴ difference 1 *
53 ䷴ to 54 ䷵ difference 6 *****
54 ䷵ to 55 ䷶ difference 2 **
55 ䷶ to 56 ䷷ difference 2 **
56 ䷷ to 57 ䷸ difference 3 ***
57 ䷸ to 58 ䷹ difference 4 ****
58 ䷹ to 59 ䷺ difference 3 ***
59 ䷺ to 60 ䷻ difference 2 **
60 ䷻ to 61 ䷼ difference 1 *
61 ䷼ to 62 ䷽ difference 6 *****
62 ䷽ to 63 ䷾ difference 3 ***
63 ䷾ to 64 ䷿ difference 6 *****

0 line changes total 0
1 line changes total 2
2 line changes total 20
3 line changes total 13
4 line changes total 19

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5 line changes total 0
6 line changes total 9
```

Spark line:



```
--- Wald-Wolfowitz runs test ---
```

Tests whether the wave values are randomly ordered (vs. clustered or alternating).

Values split at median (3): 28 above, 22 below (13 excluded ties)

Observed runs: 33, expected: 25.6, $Z = +2.13$

Significant at 95% level ($p = 0.033$): the wave shows alternation.

Note: does not survive Bonferroni correction for 28 tests (threshold $p < 0.0018$).

Difference wave bar chart

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Difference wave bar chart

A visual representation of the first-order difference wave. Each column shows how many lines changed in that transition. The horizontal axis is the position in the King Wen sequence (1-63), the vertical axis is the number of lines changed (1-6). This makes the wave pattern easier to see than a list of numbers.

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[illegible]

Trigram analysis

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Trigram analysis

Each hexagram is built from two trigrams (3-line figures): an upper and a lower. There are 8 possible trigrams. This section counts how often each trigram appears in each position, checks how often the upper or lower trigram changes between consecutive hexagrams, and shows the full 8x8 transition matrix (which trigram follows which). A uniform distribution (8 of each) confirms that all 64 possible upper/lower combinations are used exactly once.

Trigram	Upper	Lower	Total
---------	-------	-------	-------

Kun	Earth	8	8	16
-----	-------	---	---	----

Zhen Thunder	8	8	16
--------------	---	---	----

Kan	Water	8	8	16
-----	-------	---	---	----

Dui Lake	8	8	16
----------	---	---	----

Gen	Mountain	8	8	16
-----	----------	---	---	----

Li	Fire	8	8	16
----	------	---	---	----

Xun	Wind	8	8	16
-----	------	---	---	----

Qian Heaven	8	8	16
-------------	---	---	----

```

--- Consecutive trigram transitions ---

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Upper trigram changes: 59/63 transitions (93.7%)

Lower trigram changes: 58/63 transitions (92.1%)

```
--- Upper trigram transition matrix ---
```

Rows = from, Columns = to

Kun Zhen Kan Dui Gen Li Xun Qian

Kun 0 1 2 1 0 0 2 2

Zhen 0 1 1 1 2 2 0 1

Kan	0	1	0	1	1	2	2	1
-----	---	---	---	---	---	---	---	---

Dui 1 1 2 0 1 1 1 1

Gen	2	0	1	1	2	0	2	0
-----	---	---	---	---	---	---	---	---

Li 2 1 1 1 1 0 1 0

Xun	0	2	1	2	0	2	0	1
-----	---	---	---	---	---	---	---	---

Qian	3	1	0	1	1	1	0	1
------	---	---	---	---	---	---	---	---

```
--- Lower trigram transition matrix ---
Rows = from, Columns = to
      Kun   Zhen Kan   Dui   Gen   Li   Xun   Qian
Kun    0     4    0     0     0    2    1    1
Zhen   0     1    1     0     1    1    2    2
Kan    1     0    1     2     0    1    1    1
Dui    1     1    1     1     2    1    0    1
Gen    1     0    1     1     1    1    2    1
Li     1     0    1     1     2    1    1    1
Xun    1     1    2     2     1    1    0    0
Qian   3     1    1     1     1    0    1    0

--- Trigram change rates vs null (pair-preserving permutation test) ---
Null: 10,000 orderings preserving the pair structure (permute pairs + flip orientations),
the correct baseline per CRITIQUE.md. Statistic: upper/lower trigram change counts.
KW upper changes 59/63: percentile 47.3 (fraction of null <= KW)
KW lower changes 58/63: percentile 27.3

--- Pure (doubled-trigram) hexagram placement ---
The 8 pure hexagrams (upper == lower trigram). Lai Zhide (1525-1604, via Schulz 1982)
observed kan/li doubles closing both Classics; measured here against the same null.
Pure hexagram positions (1-based): [1, 2, 29, 30, 51, 52, 57, 58]
Pure hexagrams at Classic ends (positions 1,2,29,30,63,64): 4 of a possible 6
Null P(>= KW's 4) = 0.0336 (pair-preserving null; note KW's C4 fixes 1,2 by
definition, so interpret against the constrained baseline)

--- Nuclear trigram structure ---
Nuclear hexagram = lines 2-4 (lower nuclear) + 3-5 (upper nuclear). Classical fact
(commentary tradition): iterating the nuclear map sends all 64 hexagrams to 16, then to
the 4 fixed points 63, 0, and the two alternators 21(010101b rendering varies), 42.
Distinct after 1 application: 16; after 2: 4; after 3: 4 -> [0, 21, 42, 63]
KW nuclear-hexagram changes along the sequence: 58/63

--- Jing Fang Eight-Palaces rank correlation ---
Spearman rank correlation between KW positions and Jing Fang's palace ordering
(the trigram-generated classical alternative; ordering as in solve.c --null-historical).
Spearman rho = 0.1384; null P(|rho| >= observed) = 0.1170 (pair-preserving null)

--- Symmetry group vs the trigram decomposition (ROAE SYMMETRY_SEARCH theorem) ---
The C1-C5 symmetry group B3 (order 48) acts on LINE positions; reversal maps the upper
trigram to the reversed lower trigram, so most of the group does NOT preserve the
upper/lower split. Preserving elements = permutations acting within {0,1,2} and {3,4,5}
blocks (or swapping them wholesale) that also centralize rev.
Split-respecting subgroup order: 12 of 48 (computed over the rev-centralizer)

--- Methodological note ---
With ~1 expected observation per cell, no goodness-of-fit test (e.g.,
chi-square) has sufficient power to detect deviations from uniform
transitions. The matrices are descriptive only.
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Nuclear hexagram analysis

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Nuclear hexagram analysis
Every hexagram contains a hidden 'nuclear' hexagram inside it, formed by
extracting the 4 middle lines (lines 2-3-4-5). Lines 2-3-4 form the lower
nuclear trigram, and lines 3-4-5 form the upper nuclear trigram (sharing lines
3 and 4). This creates a chain: each hexagram points to its nuclear hexagram,
which points to its own nuclear hexagram, and so on until the chain loops.
These chains reveal deep structural relationships between hexagrams that are
not obvious from the surface arrangement.
---
01 ䷁ nuclear: 01 ䷁ chain: 01 ䷁ -> 01 ䷁
02 ䷇ nuclear: 02 ䷇ chain: 02 ䷇ -> 02 ䷇
03 ䷊ nuclear: 23 ䷊ chain: 03 ䷊ -> 23 ䷊ -> 02 ䷇ -> 02 ䷇
04 ䷋ nuclear: 24 ䷋ chain: 04 ䷋ -> 24 ䷋ -> 02 ䷇ -> 02 ䷇
05 ䷌ nuclear: 38 ䷌ chain: 05 ䷌ -> 38 ䷌ -> 63 ䷌ -> 64 ䷌ -> 63 ䷌
06 ䷍ nuclear: 37 ䷍ chain: 06 ䷍ -> 37 ䷍ -> 64 ䷍ -> 63 ䷌ -> 64 ䷌
07 ䷎ nuclear: 24 ䷎ chain: 07 ䷎ -> 24 ䷎ -> 02 ䷇ -> 02 ䷇
08 ䷏ nuclear: 23 ䷏ chain: 08 ䷏ -> 23 ䷏ -> 02 ䷇ -> 02 ䷇
09 ䷐ nuclear: 38 ䷐ chain: 09 ䷐ -> 38 ䷐ -> 63 ䷌ -> 64 ䷌ -> 63 ䷌
```

10	䷛ nuclear: 37	䷛ chain: 10	䷛ -> 37	䷛ -> 64	䷛ -> 63	䷛ -> 64
11	䷛ nuclear: 54	䷛ chain: 11	䷛ -> 54	䷛ -> 63	䷛ -> 64	䷛ -> 63
12	䷛ nuclear: 53	䷛ chain: 12	䷛ -> 53	䷛ -> 64	䷛ -> 63	䷛ -> 64
13	䷛ nuclear: 44	䷛ chain: 13	䷛ -> 44	䷛ -> 01	䷛ -> 01	䷛
14	䷛ nuclear: 43	䷛ chain: 14	䷛ -> 43	䷛ -> 01	䷛ -> 01	䷛
15	䷛ nuclear: 40	䷛ chain: 15	䷛ -> 40	䷛ -> 63	䷛ -> 64	䷛ -> 63
16	䷛ nuclear: 39	䷛ chain: 16	䷛ -> 39	䷛ -> 64	䷛ -> 63	䷛ -> 64
17	䷛ nuclear: 53	䷛ chain: 17	䷛ -> 53	䷛ -> 64	䷛ -> 63	䷛ -> 64
18	䷛ nuclear: 54	䷛ chain: 18	䷛ -> 54	䷛ -> 63	䷛ -> 64	䷛ -> 63
19	䷛ nuclear: 24	䷛ chain: 19	䷛ -> 24	䷛ -> 02	䷛ -> 02	䷛
20	䷛ nuclear: 23	䷛ chain: 20	䷛ -> 23	䷛ -> 02	䷛ -> 02	䷛
21	䷛ nuclear: 39	䷛ chain: 21	䷛ -> 39	䷛ -> 64	䷛ -> 63	䷛ -> 64
22	䷛ nuclear: 40	䷛ chain: 22	䷛ -> 40	䷛ -> 63	䷛ -> 64	䷛ -> 63
23	䷛ nuclear: 02	䷛ chain: 23	䷛ -> 02	䷛ -> 02	䷛	䷛
24	䷛ nuclear: 02	䷛ chain: 24	䷛ -> 02	䷛ -> 02	䷛	䷛
25	䷛ nuclear: 53	䷛ chain: 25	䷛ -> 53	䷛ -> 64	䷛ -> 63	䷛ -> 64
26	䷛ nuclear: 54	䷛ chain: 26	䷛ -> 54	䷛ -> 63	䷛ -> 64	䷛ -> 63
27	䷛ nuclear: 02	䷛ chain: 27	䷛ -> 02	䷛ -> 02	䷛	䷛
28	䷛ nuclear: 01	䷛ chain: 28	䷛ -> 01	䷛ -> 01	䷛	䷛
29	䷛ nuclear: 27	䷛ chain: 29	䷛ -> 27	䷛ -> 02	䷛ -> 02	䷛
30	䷛ nuclear: 28	䷛ chain: 30	䷛ -> 28	䷛ -> 01	䷛ -> 01	䷛
31	䷛ nuclear: 44	䷛ chain: 31	䷛ -> 44	䷛ -> 01	䷛ -> 01	䷛
32	䷛ nuclear: 43	䷛ chain: 32	䷛ -> 43	䷛ -> 01	䷛ -> 01	䷛
33	䷛ nuclear: 44	䷛ chain: 33	䷛ -> 44	䷛ -> 01	䷛ -> 01	䷛
34	䷛ nuclear: 43	䷛ chain: 34	䷛ -> 43	䷛ -> 01	䷛ -> 01	䷛
35	䷛ nuclear: 39	䷛ chain: 35	䷛ -> 39	䷛ -> 64	䷛ -> 63	䷛ -> 64
36	䷛ nuclear: 40	䷛ chain: 36	䷛ -> 40	䷛ -> 63	䷛ -> 64	䷛ -> 63
37	䷛ nuclear: 64	䷛ chain: 37	䷛ -> 64	䷛ -> 63	䷛ -> 64	䷛
38	䷛ nuclear: 63	䷛ chain: 38	䷛ -> 63	䷛ -> 64	䷛ -> 63	䷛
39	䷛ nuclear: 64	䷛ chain: 39	䷛ -> 64	䷛ -> 63	䷛ -> 64	䷛
40	䷛ nuclear: 63	䷛ chain: 40	䷛ -> 63	䷛ -> 64	䷛ -> 63	䷛
41	䷛ nuclear: 24	䷛ chain: 41	䷛ -> 24	䷛ -> 02	䷛ -> 02	䷛
42	䷛ nuclear: 23	䷛ chain: 42	䷛ -> 23	䷛ -> 02	䷛ -> 02	䷛
43	䷛ nuclear: 01	䷛ chain: 43	䷛ -> 01	䷛ -> 01	䷛	䷛
44	䷛ nuclear: 01	䷛ chain: 44	䷛ -> 01	䷛ -> 01	䷛	䷛
45	䷛ nuclear: 53	䷛ chain: 45	䷛ -> 53	䷛ -> 64	䷛ -> 63	䷛ -> 64
46	䷛ nuclear: 54	䷛ chain: 46	䷛ -> 54	䷛ -> 63	䷛ -> 64	䷛ -> 63
47	䷛ nuclear: 37	䷛ chain: 47	䷛ -> 37	䷛ -> 64	䷛ -> 63	䷛ -> 64
48	䷛ nuclear: 38	䷛ chain: 48	䷛ -> 38	䷛ -> 63	䷛ -> 64	䷛ -> 63
49	䷛ nuclear: 44	䷛ chain: 49	䷛ -> 44	䷛ -> 01	䷛ -> 01	䷛
50	䷛ nuclear: 43	䷛ chain: 50	䷛ -> 43	䷛ -> 01	䷛ -> 01	䷛
51	䷛ nuclear: 39	䷛ chain: 51	䷛ -> 39	䷛ -> 64	䷛ -> 63	䷛ -> 64
52	䷛ nuclear: 40	䷛ chain: 52	䷛ -> 40	䷛ -> 63	䷛ -> 64	䷛ -> 63
53	䷛ nuclear: 64	䷛ chain: 53	䷛ -> 64	䷛ -> 63	䷛ -> 64	䷛
54	䷛ nuclear: 63	䷛ chain: 54	䷛ -> 63	䷛ -> 64	䷛ -> 63	䷛
55	䷛ nuclear: 28	䷛ chain: 55	䷛ -> 28	䷛ -> 01	䷛ -> 01	䷛
56	䷛ nuclear: 28	䷛ chain: 56	䷛ -> 28	䷛ -> 01	䷛ -> 01	䷛
57	䷛ nuclear: 38	䷛ chain: 57	䷛ -> 38	䷛ -> 63	䷛ -> 64	䷛ -> 63
58	䷛ nuclear: 37	䷛ chain: 58	䷛ -> 37	䷛ -> 64	䷛ -> 63	䷛ -> 64
59	䷛ nuclear: 27	䷛ chain: 59	䷛ -> 27	䷛ -> 02	䷛ -> 02	䷛
60	䷛ nuclear: 27	䷛ chain: 60	䷛ -> 27	䷛ -> 02	䷛ -> 02	䷛
61	䷛ nuclear: 27	䷛ chain: 61	䷛ -> 27	䷛ -> 02	䷛ -> 02	䷛
62	䷛ nuclear: 28	䷛ chain: 62	䷛ -> 28	䷛ -> 01	䷛ -> 01	䷛
63	䷛ nuclear: 64	䷛ chain: 63	䷛ -> 64	䷛ -> 63	䷛	䷛
64	䷛ nuclear: 63	䷛ chain: 64	䷛ -> 63	䷛ -> 64	䷛	䷛

--- Nuclear hexagram frequency ---

How often each hexagram appears as another's nuclear hexagram

01	䷁ The Creative	appears 4 times
02	䷁ The Receptive	appears 4 times
23	䷧ Splitting Apart	appears 4 times
24	䷄ Return	appears 4 times
27	䷴ Nourishment	appears 4 times
28	䷌ Great Preponderance	appears 4 times
37	䷤ The Family	appears 4 times
38	䷧ Opposition	appears 4 times
39	䷧ Obstruction	appears 4 times
40	䷌ Deliverance	appears 4 times
43	䷌ Breakthrough	appears 4 times
44	䷌ Coming to Meet	appears 4 times
53	䷌ Development	appears 4 times
54	䷌ The Marrying Maiden	appears 4 times
63	䷌ After Completion	appears 4 times
64	䷌ Before Completion	appears 4 times

```
--- Null model note ---
Nuclear hexagram derivation is a fixed function of binary value (lines 2-5),
independent of the King Wen ordering. The chain structure, cycle lengths,
and frequency distribution above are identical for ANY ordering of the 64
hexagrams. These properties reflect the 6-bit binary system, not the
King Wen sequence specifically.
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Line change positional analysis

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Line change positional analysis
Each hexagram has 6 lines (positions 1-6, bottom to top). When consecutive
hexagrams differ, which specific lines are changing? If the King Wen sequence
were random, each line position would change about 50% of the time. Significant
deviation from 50% would suggest the ordering favors changing certain positions.
The co-change matrix shows which pairs of lines tend to flip together in the
same transition – revealing correlated line movements.
---
Line 1: 37/63 ( 58.7%) #####
Line 2: 32/63 ( 50.8%) #####
Line 3: 35/63 ( 55.6%) #####
Line 4: 34/63 ( 54.0%) #####
Line 5: 35/63 ( 55.6%) #####
Line 6: 38/63 ( 60.3%) #####
```

```
--- Line co-change frequency ---
How often pairs of lines change together in the same transition
```

	L1	L2	L3	L4	L5	L6
L1	.	18	20	19	19	30
L2	.	.	20	17	23	20
L3	.	.	.	25	16	21
L4	.	.	.	.	20	20
L5	.	.	.	.	.	20
L6	.	.	.	.	.	.

Complement distance analysis

```
---
Complement distance analysis
Every hexagram has an 'opposite' (complement) formed by toggling all 6 lines
(solid becomes broken, broken becomes solid). For example, The Creative (all
solid) and The Receptive (all broken) are complements. This section finds where
each hexagram's complement sits in the King Wen sequence and how far apart they
are. If complements tend to be near each other, it suggests the sequence was
deliberately organized around opposition; if far apart, they may serve as
bookends for larger structural sections.
---
01 ䷀ <-> 02 ䷁ distance: 1 The Creative <-> The Receptive
02 ䷁ <-> 01 ䷀ distance: 1 The Receptive <-> The Creative
03 ䷂ <-> 50 ䷌ distance: 47 Difficulty at the Beginning <-> The Cauldron
04 ䷃ <-> 49 ䷋ distance: 45 Youthful Folly <-> Revolution
05 ䷄ <-> 35 ䷍ distance: 30 Waiting <-> Progress
06 ䷅ <-> 36 ䷄ distance: 30 Conflict <-> Darkening of the Light
07 ䷆ <-> 13 ䷗ distance: 6 The Army <-> Fellowship
08 ䷇ <-> 14 ䷔ distance: 6 Holding Together <-> Great Possession
09 ䷈ <-> 16 ䷖ distance: 7 Small Taming <-> Enthusiasm
10 ䷉ <-> 15 ䷜ distance: 5 Treading <-> Modesty
11 ䷊ <-> 12 ䷌ distance: 1 Peace <-> Standstill
12 ䷌ <-> 11 ䷊ distance: 1 Standstill <-> Peace
13 ䷗ <-> 07 ䷆ distance: 6 Fellowship <-> The Army
14 ䷔ <-> 08 ䷇ distance: 6 Great Possession <-> Holding Together
15 ䷜ <-> 10 ䷉ distance: 5 Modesty <-> Treading
16 ䷖ <-> 09 ䷈ distance: 7 Enthusiasm <-> Small Taming
17 ䷍ <-> 18 ䷘ distance: 1 Following <-> Work on the Decayed
18 ䷘ <-> 17 ䷍ distance: 1 Work on the Decayed <-> Following
19 ䷙ <-> 33 ䷯ distance: 14 Approach <-> Retreat
20 ䷚ <-> 34 ䷔ distance: 14 Contemplation <-> Great Power
21 ䷛ <-> 48 ䷛ distance: 27 Biting Through <-> The Well
```

```
22 ䷮ <-> 47 ䷮ distance: 25 Grace <-> Oppression
23 ䷮ <-> 43 ䷮ distance: 20 Splitting Apart <-> Breakthrough
24 ䷮ <-> 44 ䷮ distance: 20 Return <-> Coming to Meet
25 ䷮ <-> 46 ䷮ distance: 21 Innocence <-> Pushing Upward
26 ䷮ <-> 45 ䷮ distance: 19 Great Taming <-> Gathering Together
27 ䷮ <-> 28 ䷮ distance: 1 Nourishment <-> Great Preponderance
28 ䷮ <-> 27 ䷮ distance: 1 Great Preponderance <-> Nourishment
29 ䷮ <-> 30 ䷮ distance: 1 The Abysmal <-> The Clinging
30 ䷮ <-> 29 ䷮ distance: 1 The Clinging <-> The Abysmal
31 ䷮ <-> 41 ䷮ distance: 10 Influence <-> Decrease
32 ䷮ <-> 42 ䷮ distance: 10 Duration <-> Increase
33 ䷮ <-> 19 ䷮ distance: 14 Retreat <-> Approach
34 ䷮ <-> 20 ䷮ distance: 14 Great Power <-> Contemplation
35 ䷮ <-> 05 ䷮ distance: 30 Progress <-> Waiting
36 ䷮ <-> 06 ䷮ distance: 30 Darkening of the Light <-> Conflict
37 ䷮ <-> 40 ䷮ distance: 3 The Family <-> Deliverance
38 ䷮ <-> 39 ䷮ distance: 1 Opposition <-> Obstruction
39 ䷮ <-> 38 ䷮ distance: 1 Obstruction <-> Opposition
40 ䷮ <-> 37 ䷮ distance: 3 Deliverance <-> The Family
41 ䷮ <-> 31 ䷮ distance: 10 Decrease <-> Influence
42 ䷮ <-> 32 ䷮ distance: 10 Increase <-> Duration
43 ䷮ <-> 23 ䷮ distance: 20 Breakthrough <-> Splitting Apart
44 ䷮ <-> 24 ䷮ distance: 20 Coming to Meet <-> Return
45 ䷮ <-> 26 ䷮ distance: 19 Gathering Together <-> Great Taming
46 ䷮ <-> 25 ䷮ distance: 21 Pushing Upward <-> Innocence
47 ䷮ <-> 22 ䷮ distance: 25 Oppression <-> Grace
48 ䷮ <-> 21 ䷮ distance: 27 The Well <-> Biting Through
49 ䷮ <-> 04 ䷮ distance: 45 Revolution <-> Youthful Folly
50 ䷮ <-> 03 ䷮ distance: 47 The Cauldron <-> Difficulty at the Beginning
51 ䷮ <-> 57 ䷮ distance: 6 The Arousing <-> The Gentle
52 ䷮ <-> 58 ䷮ distance: 6 Keeping Still <-> The Joyous
53 ䷮ <-> 54 ䷮ distance: 1 Development <-> The Marrying Maiden
54 ䷮ <-> 53 ䷮ distance: 1 The Marrying Maiden <-> Development
55 ䷮ <-> 59 ䷮ distance: 4 Abundance <-> Dispersion
56 ䷮ <-> 60 ䷮ distance: 4 The Wanderer <-> Limitation
57 ䷮ <-> 51 ䷮ distance: 6 The Gentle <-> The Arousing
58 ䷮ <-> 52 ䷮ distance: 6 The Joyous <-> Keeping Still
59 ䷮ <-> 55 ䷮ distance: 4 Dispersion <-> Abundance
60 ䷮ <-> 56 ䷮ distance: 4 Limitation <-> The Wanderer
61 ䷮ <-> 62 ䷮ distance: 1 Inner Truth <-> Small Preponderance
62 ䷮ <-> 61 ䷮ distance: 1 Small Preponderance <-> Inner Truth
63 ䷮ <-> 64 ䷮ distance: 1 After Completion <-> Before Completion
64 ䷮ <-> 63 ䷮ distance: 1 Before Completion <-> After Completion
```

--- Complement distance statistics ---

Mean distance: 12.1

Median distance: 6

Min distance: 1 (adjacent = pair is an inverse pair)

Max distance: 47

--- Complement distance null model ---

Random mean complement distance (over 10000 shuffles): 21.6

King Wen mean complement distance: 12.1

King Wen percentile vs random: 0.0%

Complements are significantly closer together than chance would predict, suggesting the sequence was deliberately organized around opposition.

Palindrome analysis of the difference wave

Palindrome analysis of the difference wave

A palindrome reads the same forwards and backwards (like 2,4,6,4,2). Finding palindromic runs in the difference wave suggests the King Wen sequence contains deliberate mirror symmetry – sections where the pattern of change rises and falls in a balanced way. Longer palindromes are less likely to occur by chance and more likely to reflect intentional design.

Found 27 palindromic subsequences (showing longest 20)

Length 7 at positions 11-17: [6,2,2,4,2,2,6] 

Length 7 at positions 19-25: [4,3,2,2,2,3,4] 

Length 7 at positions 39-45: [4,3,2,4,2,3,4]

Length 5 at positions 12-16: [2,2,4,2,2]

Length 5 at positions 20-24: [3,2,2,2,3]

Length 5 at positions 40-44: [3,2,4,2,3]

Length 5 at positions 43-47: [2,3,4,3,2]

Length 5 at positions 45-49: [4,3,2,3,4]

Length 5 at positions 55-59: [2,3,4,3,2]

Length 3 at positions 03-05: [4,4,4]

Length 3 at positions 07-09: [2,4,2]

Length 3 at positions 08-10: [4,2,4]

Length 3 at positions 13-15: [2,4,2]

Length 3 at positions 18-20: [3,4,3]

Length 3 at positions 21-23: [2,2,2]

Length 3 at positions 26-28: [2,6,2]

Length 3 at positions 27-29: [6,2,6]

Length 3 at positions 30-32: [3,2,3]

Length 3 at positions 33-35: [4,4,4]

Length 3 at positions 35-37: [4,2,4]

--- Palindrome count by length ---

Length 7: 3 found

Length 5: 6 found

Length 3: 18 found

--- Palindrome null model ---

How many palindromes of length >= 3 do random permutations produce?

King Wen palindromes: 27, longest: 7

Random mean palindromes: 22.0, mean longest: 6.8

King Wen palindrome count percentile: 78.7%

King Wen longest palindrome percentile: 43.8%

--- Pair-constrained palindrome null model ---

The unconstrained comparison shuffles all 64 hexagrams freely. A fairer comparison preserves the pair structure: each pair of hexagrams stays adjacent, only the pair order and within-pair orientation are randomized.

King Wen palindromes: 27, longest: 7

Pair-constrained mean palindromes: 27.4, mean longest: 8.5

King Wen palindrome count percentile (pair-constrained): 49.8%

King Wen longest palindrome percentile (pair-constrained): 13.3%

Upper Canon (1-30) vs. Lower Canon (31-64) comparison

Upper Canon (1-30) vs. Lower Canon (31-64) comparison

The I Ching is traditionally divided into two books: the Upper Canon (hexagrams 1-30) and Lower Canon (hexagrams 31-64). This division is ancient and may reflect different authorship, time periods, or thematic groupings. This section compares the mathematical properties of each half: do they have similar difference wave patterns? Similar yin/yang balance? Does the 'no 5-line transition' property hold in each half independently, or only in the whole?

Metric Upper (1-30) Lower (31-64)

Hexagrams	30	34
Transitions	29	33
Mean line changes	3.38	3.33
Yang lines (total)	86	106
Yin lines (total)	94	98

Difference distribution: Upper (1-30) Lower (31-64)

0 line changes	0	0
1 line changes	0	2
2 line changes	12	8
3 line changes	4	8
4 line changes	8	11
5 line changes	0	0
6 line changes	5	4

Upper Canon wave:

Lower Canon wave:

5-line transitions in Upper Canon: 0
5-line transitions in Lower Canon: 0

--- Canon split null model ---
Permutation test: is the King Wen split at position 30 special, or would any random split of the 64-hexagram sequence show a similar gap in mean line-change differences between the two halves?
King Wen |upper_mean - lower_mean|: 0.0460
Random permutations with gap >= King Wen: 8806/10000
King Wen gap percentile: 11.9%
The King Wen canon split does not produce a statistically significant mean-difference gap. The observed difference between upper and lower halves is consistent with what random shuffles produce.

Autocorrelation of the difference wave

Autocorrelation of the difference wave
Autocorrelation measures whether the wave is correlated with a shifted copy of itself. A peak at lag N means the pattern tends to repeat every N steps. A value near +1 means strong repetition, near -1 means strong alternation, and near 0 means no relationship. If the King Wen ordering contains hidden periodic structure, it will show up as peaks in this analysis.
Caveat: with only N=63 data points, the 95% noise threshold is +/-0.25. Values within that band are indistinguishable from random noise.

Mean difference: 3.349
Variance: 1.910
95% noise threshold: +/-0.247 (values inside this are not significant)

Lag	Autocorrelation	Sig?	Visualization
0	1.0000		[: : #]
1	-0.2469		[# :]
2	-0.0098		[: :]
3	-0.0491		[: :]
4	-0.0447		[: :]
5	-0.0457		[: :]
6	-0.0218		[: :]
7	-0.0644		[: :]
8	0.1204		[: # :]
9	-0.1275		[: # :]
10	0.1375		[: #:]
11	-0.0290		[: :]
12	-0.0221		[: :]
13	0.0850		[: # :]
14	-0.1596		[:# :]
15	-0.0110		[: :]
16	0.1015		[: # :]
17	-0.1514		[:# :]
18	0.0055		[: :]
19	0.0851		[: # :]
20	-0.1403		[:# :]
21	0.0274		[: :]
22	-0.1315		[: # :]
23	-0.0137		[: :]
24	0.2263		[: #]
25	-0.1595		[:# :]
26	0.1112		[: # :]
27	-0.1109		[: # :]
28	0.1649		[: #:]
29	0.0093		[: :]
30	-0.0969		[: # :]
31	-0.0480		[: :]

Significant lags (outside noise band): 0/31
(Expected false positives at 95% threshold with 31 lags: ~1.6)
No significant autocorrelation detected – consistent with no hidden periodicity, but note that N=63 provides limited statistical power to detect weak periodicity.

Spectral analysis (Discrete Fourier Transform)

Spectral analysis (Discrete Fourier Transform)
The DFT decomposes the difference wave into frequency components, like splitting white light into a rainbow. Each frequency represents a periodic pattern in the wave. A strong peak at frequency F means the wave has a repeating cycle every 63/F transitions. If the King Wen ordering has hidden periodic structure, the spectrum will show clear peaks rather than flat noise.
Caveat: with only 63 samples, each of the 31 frequency bins has wide uncertainty. A flat spectrum may indicate no periodicity, or insufficient data.

Number of samples: 63
Mean (DC component): 3.349
White noise floor: 0.1741 (magnitudes near this are not significant)

Freq	Period	Magnitude	Sig?	Spectrum
1	63.0	0.0224		##
2	31.5	0.1823		#####
3	21.0	0.0145		#
4	15.8	0.1359		#####
5	12.6	0.2021		#####
6	10.5	0.1157		#####
7	9.0	0.2399		#####
8	7.9	0.1092		#####
9	7.0	0.0553		####
10	6.3	0.1630		#####
11	5.7	0.2371		#####
12	5.2	0.0473		####
13	4.8	0.2326		#####
14	4.5	0.0331		###
15	4.2	0.1145		#####
16	3.9	0.2204		#####
17	3.7	0.1451		#####
18	3.5	0.1975		#####
19	3.3	0.2197		#####
20	3.1	0.2166		#####
21	3.0	0.1611		#####
22	2.9	0.0222		##
23	2.7	0.1210		#####
24	2.6	0.4266	*	#####
25	2.5	0.0306		##
26	2.4	0.2297		#####
27	2.3	0.0795		#####
28	2.2	0.0639		####
29	2.2	0.2114		#####
30	2.1	0.2129		#####
31	2.0	0.2033		#####

Frequencies above 2x noise floor: 1/31
(The 2x threshold is ad hoc. A proper test would use Fisher's g-statistic or Bonferroni correction across frequency bins. With only 63 samples, even real periodicity may not rise above the noise floor.)

Markov chain analysis

Markov chain analysis
If we treat each difference value as a 'state', we can ask: does the current difference predict the next one? In a random sequence, knowing that the last transition changed 2 lines tells you nothing about the next transition. But if certain patterns like '6 is always followed by 2' emerge, it reveals hidden rules in the King Wen ordering. The matrix shows the probability of each transition between difference values.

Transition probability matrix (row = from, column = to)
-> 1 -> 2 -> 3 -> 4 -> 6
1 0.00 0.00 0.00 0.00 1.00 (n=2)

2	0.05	0.25	0.25	0.30	0.15	(n=20)
3	0.00	0.46	0.00	0.46	0.08	(n=13)
4	0.05	0.26	0.26	0.32	0.11	(n=19)
6	0.00	0.50	0.38	0.12	0.00	(n=8)

```

--- Notable transition patterns ---
(Caveat: rows with small n are unreliable – patterns based on <5 observations
are anecdotes, not statistically meaningful.)
3 -> 2: 46% (6/13 times)
3 -> 4: 46% (6/13 times)

--- Permutation test ---
Is the transition matrix more concentrated than random orderings produce?
King Wen matrix concentration: 2.3303
Random orderings equally or more concentrated: 2221/5000 (44.4%)

```

Gray code comparison

```

---
Gray code comparison
A Gray code is an ordering of binary values where each consecutive pair differs
by exactly 1 bit. For 6-bit hexagrams, a Gray code path would change exactly 1
line at every step – the smoothest possible sequence. The King Wen sequence does
NOT follow a Gray code (most transitions change 2-6 lines). This comparison
shows how much 'rougher' King Wen is than the theoretical minimum, suggesting
the ordering prioritizes something other than smooth transitions.
---

```

Metric	Gray Code	King Wen
Total path length	63	211
Mean change per transition	1.00	3.35
Max single transition	1	6
Min single transition	1	1
King Wen / Gray Code ratio		3.35x

Difference distribution:	Gray Code	King Wen
1 line changes	63	2
2 line changes	0	20
3 line changes	0	13
4 line changes	0	19
6 line changes	0	9


```

Gray code wave:
King Wen wave:

```



```

--- Methodological note ---
This comparison is descriptive. The Gray code ratio (King Wen path / Gray
code path) is not a statistical test. For significance testing of King Wen's
path length against appropriate null models, see --path.

```

Symmetry group analysis (XOR algebra)

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---
Symmetry group analysis (XOR algebra)
The 64 hexagrams form a mathematical group under XOR: combining any two hexagrams
by toggling their differing lines always produces another valid hexagram. This is
like mixing colors – every combination is itself a color. Under this algebra,
certain sets of hexagrams form 'subgroups' (closed sets where combining any two
members stays in the set). This section checks whether the King Wen pairs, inverse
pairs, and the canonical divisions correspond to meaningful algebraic subgroups.
---
XOR identity element: 0b000000 (hexagram 2, The Receptive)

--- Fixed points under complement (XOR 0b111111) ---
Hexagrams that are their own complement do not exist (since XOR with
0b111111 always flips all bits, no 6-bit value equals its own complement).

--- King Wen pair XOR products ---

```

For each pair, XOR the two hexagrams to see what 'difference element' they produce.
If many pairs produce the same XOR product, the pairing has algebraic regularity.

XOR Product	Hexagram	Pair count	Pairs
001100 62 ䷋	4	9-10, 15-16, 21-22, 47-48	
010010 29 ䷔	4	7-8, 13-14, 31-32, 41-42	
011110 28 ䷌	4	25-26, 37-38, 39-40, 45-46	
100001 27 ䷌	4	23-24, 43-44, 55-56, 59-60	
101101 30 ䷌	4	5-6, 35-36, 51-52, 57-58	
110011 61 ䷌	4	3-4, 19-20, 33-34, 49-50	
111111 01 ䷌	8	1-2, 11-12, 17-18, 27-28, 29-30, 53-54, 61-62, 63-64	

Unique XOR products: 7 (out of 32 pairs)
If this number is small, the pairs share algebraic structure.

--- Inverse pair XOR closure test ---
Inverse pair hexagrams (16 total): 01, 02, 11, 12, 17, 18, 27, 28, 29, 30, 53, 54, 61, 62, 63, 64
Closed under XOR: Yes – forms a subgroup

Alternative sequence comparison

Alternative sequence comparison
The King Wen ordering is not the only way to arrange 64 hexagrams. The Fu Xi (binary) sequence orders them by numerical value (0-63), which is mathematically natural but has no traditional significance. The Mawangdui sequence was found on silk manuscripts in a 168 BCE tomb and may represent an independent tradition. Comparing the same analyses across orderings reveals what is unique to King Wen.

Metric	King Wen	Fu Xi (binary)	Mawangdui
Total path length	211	120	104
Mean change	3.35	1.90	1.65
1-line transitions	2	32	32
2-line transitions	20	16	24
3-line transitions	13	8	4
4-line transitions	19	4	3
5-line transitions	0	2	0
6-line transitions	9	1	0



--- What makes King Wen unique ---
Zero 5-line transitions: King Wen=0, Fu Xi=2, Mawangdui=0
Zero 0-line transitions: King Wen=0, Fu Xi=0, Mawangdui=0

Windowed entropy analysis

Windowed entropy analysis
Instead of one entropy value for the whole wave, we slide a window across it and compute entropy locally. This reveals WHERE in the sequence structure concentrates (low entropy = predictable) vs. dissolves (high entropy = varied). Dips in the curve mark regions with repetitive transition patterns.

Window size: 15		
Center	Entropy	Visualization
8	1.7056	#####
9	1.5628	#####
10	1.7056	#####
11	1.8323	#####
12	1.8323	#####
13	1.8892	#####
14	1.7968	#####

```
15 1.7968 #####
16 1.7232 #####
17 1.8295 #####
18 1.8295 #####
19 1.6729 #####
20 1.8295 #####
21 1.8295 #####
22 1.8295 #####
23 1.8892 #####
24 1.8892 #####
25 1.8323 #####
26 1.8892 #####
27 1.8892 #####
28 1.8892 #####
29 1.8892 #####
30 1.9086 #####
31 1.9656 #####
32 1.8892 #####
33 1.9656 #####
34 1.9656 #####
35 1.8892 #####
36 1.8892 #####
37 1.8062 #####
38 1.7465 #####
39 1.7465 #####
40 1.7465 #####
41 1.8062 #####
42 1.8062 #####
43 1.8062 #####
44 1.7465 #####
45 2.0226 #####
46 2.0226 #####
47 2.0662 #####
48 2.0419 #####
49 2.0662 #####
50 2.0662 #####
51 2.0419 #####
52 2.0662 #####
53 2.1736 #####
54 2.2566 #####
55 2.2566 #####
56 2.2892 #####
```

Mean windowed entropy: 1.8956
Min entropy: 1.5628 at position 9 (most structured)
Max entropy: 2.2892 at position 56 (most varied)

Entropy spark: 

--- Methodological note ---
Windowed entropy is an exploratory visualization, not a statistical test.
Apparent patterns (dips and peaks) are expected from random variation in a short sequence. Without a null model, no significance can be assigned to specific regions. Interpret as descriptive, not inferential.

Mutual information: upper vs. lower trigram transitions

Mutual information: upper vs. lower trigram transitions
When two consecutive hexagrams differ, do the upper and lower trigrams change independently, or is knowing that the upper trigram changed informative about whether the lower trigram also changed? Mutual information quantifies this:
0 bits = completely independent (knowing one tells you nothing about the other)
higher = correlated (they tend to change together or stay together)

Joint distribution of trigram changes:

	Lower unchanged	Lower changed
Upper unchanged	0 (0.0%)	4 (6.3%)
Upper changed	5 (7.9%)	54 (85.7%)

Mutual information: 0.0078 bits

Normalized MI: 0.0230 (0=independent, 1=perfectly correlated)

Mean MI of random permutations: 0.0203 bits
King Wen percentile: 7.6%

--- Full 8-state trigram mutual information ---
Using actual trigram identities (8 possible values each) rather than binary changed/unchanged, which preserves more information.
MI(upper, lower) across all 64 hexagrams: 0.000000 bits
(Expected: 0.0 – since all 64 combinations appear exactly once, the trigrams are perfectly independent in the static set.)
Note: with all 64 upper/lower trigram combinations present exactly once, independence is expected by construction (complete Latin square).

Yin-yang balance wave

Yin-yang balance wave
Each hexagram has 6 lines that are either yang (solid, 1) or yin (broken, 0).
Note: since the King Wen sequence contains all 64 possible hexagrams exactly once, the total yin-yang count is necessarily balanced (192 each). This is a mathematical tautology, not a property of the ordering. However, the LOCAL balance as you move through the sequence IS a property of the ordering – it shows where yang-dominant or yin-dominant hexagrams cluster together.

Total yang lines: 192 / 384 (50.0%)
Total yin lines: 192 / 384 (50.0%)
Mean yang per hexagram: 3.00

Running yang average (window=8):				
Pos	Yang	RunAvg	Visualization	
1	6	6.00	[	#]
2	0	3.00	[	#]
3	2	2.67	[	#]
4	2	2.50	[	#]
5	4	2.80	[	#]
6	4	3.00	[	#]
7	1	2.71	[	#]
8	1	2.50	[	#]
9	5	2.38	[	#]
10	5	3.00	[	#]
11	3	3.12	[	#]
12	3	3.25	[	#]
13	5	3.38	[	#]
14	5	3.50	[	#]
15	1	3.50	[	#]
16	1	3.50	[	#]
17	3	3.25	[	#]
18	3	3.00	[	#]
19	2	2.88	[	#]
20	2	2.75	[	#]
21	3	2.50	[	#]
22	3	2.25	[	#]
23	1	2.25	[	#]
24	1	2.25	[	#]
25	4	2.38	[	#]
26	4	2.50	[	#]
27	2	2.50	[	#]
28	4	2.75	[	#]
29	2	2.62	[	#]
30	4	2.75	[	#]
31	3	3.00	[	#]
32	3	3.25	[	#]
33	4	3.25	[	#]
34	4	3.25	[	#]
35	2	3.25	[	#]
36	2	3.00	[	#]
37	4	3.25	[	#]
38	4	3.25	[	#]
39	2	3.12	[	#]
40	2	3.00	[	#]

41	3	2.88	[	#	]
42	3	2.75	[	#	]
43	5	3.12	[	#	]
44	5	3.50	[	#	]
45	2	3.25	[	#	]
46	2	3.00	[	#	]
47	3	3.12	[	#	]
48	3	3.25	[	#	]
49	4	3.38	[	#	]
50	4	3.50	[	#	]
51	2	3.12	[	#	]
52	2	2.75	[	#	]
53	3	2.88	[	#	]
54	3	3.00	[	#	]
55	3	3.00	[	#	]
56	3	3.00	[	#	]
57	4	3.00	[	#	]
58	4	3.00	[	#	]
59	3	3.12	[	#	]
60	3	3.25	[	#	]
61	4	3.38	[	#	]
62	2	3.25	[	#	]
63	3	3.25	[	#	]
64	3	3.25	[	#	]

Yang count spark: 

--- Yang/Yin dominance runs ---
Positions 02-04: 3 consecutive yin-dominant hexagrams

Odd-vs-even transition parity (McKenna 25/75)

Odd-vs-even transition parity (McKenna's 25/75 observation)

Each transition $d(s_i, s_{i+1})$ is between 1 and 6 line changes.
Theorem 1 says within-pair distances are always even (in $\{2,4,6\}$),
so all 32 within-pair transitions are forced even by C1. The 31
between-pair transitions can be any non-5 value.

McKenna (The Invisible Landscape, 1975) noted that ~25% of King Wen's
transitions are odd-distance and ~75% are even-distance. We report both
the LINEAR reading (the standard 63-transition difference wave) and the
CIRCULAR reading (64 transitions, including the wrap-around $s_{63} \rightarrow s_0$).

Last hexagram (position 64): #42 (101010)
First hexagram (position 1): #63 (111111)
Wrap-around distance: $\text{hamming}(\#42, \#63) = 3$ (odd)

Mode	Total	Odd	%Odd	Even	%Even
LINEAR (63 trans)	63	15	23.8095%	48	76.1905%
CIRCULAR (64 trans)	64	16	25.0000%	48	75.0000%

CIRCULAR parity is EXACTLY 25.0000% / 75.0000% – McKenna's 25/75 holds exactly
when the wrap-around transition is included. Without wrap-around, the linear
split is 23.8095% / 76.1905% – approximately but not exactly 25/75.

--- Structural note ---
All 32 within-pair transitions are forced even by C1 + Theorem 1.
So the parity split is determined entirely by the 31 (or 32 with wrap)
between-pair transitions:
Between-pair transitions: 15 odd, 16 even
Within-pair transitions: 0 odd (forced), 32 even (forced)
Plus wrap-around: 1 odd

--- 'C8' candidate: is the wrap-around parity a design feature? ---
The spec's C1-C7 do not constrain the wrap-around $s_{63} \rightarrow s_0$. So if
McKenna's exact 25/75 was an intentional design feature, it would imply
a new constraint:
C8 (weak): $\text{hamming}(s_{63}, s_0)$ is odd (admits $\{1, 3, 5\}$; C2 may extend to 64th)
C8 (strict): $\text{hamming}(s_{63}, s_0) = 3$ (King Wen's exact value)

Restrictiveness: the prior for wrap-around distance among {1,2,3,4,6} (5 forbidden by C2 if extended, 0 impossible for a permutation) is uniform → P(odd) ≈ 40%, so the weak form eliminates ~60% of completions; the strict form eliminates ~80%.

Whether C8 is a real constraint or a numerical coincidence is empirical: a Monte Carlo sample of sequences satisfying C1-C7 would tell us how non-uniform the wrap-around distribution actually is among King Wen-eligible orderings. Today this is NOT in the formal spec – it remains an observation, not a constraint.

Hexagram neighborhoods

Hexagram neighborhoods		
For each hexagram, which others are 'nearby' (differ by only 1 or 2 lines)?		
Dense neighborhoods mean a hexagram has many close relatives; sparse ones are more isolated. This also shows whether King Wen places neighbors near each other in the sequence or scatters them.		

01 ☰ The Creative	d=1: [09, 10, 13, 14, 43, 44]	(6 neighbors, mean seq dist: 21.2)
02 ☷ The Receptive	d=1: [07, 08, 15, 16, 23, 24]	(6 neighbors, mean seq dist: 13.5)
03 ☱ Difficulty at the Beginning	d=1: [08, 17, 24, 42, 60, 63]	(6 neighbors, mean seq dist: 32.7)
04 ☴ Youthful Folly	d=1: [07, 18, 23, 41, 59, 64]	(6 neighbors, mean seq dist: 31.3)
05 ☵ Waiting	d=1: [09, 11, 43, 48, 60, 63]	(6 neighbors, mean seq dist: 34.0)
06 ☲ Conflict	d=1: [10, 12, 44, 47, 59, 64]	(6 neighbors, mean seq dist: 33.3)
07 ☳ The Army	d=1: [02, 04, 19, 29, 40, 46]	(6 neighbors, mean seq dist: 19.0)
08 ☶ Holding Together	d=1: [02, 03, 20, 29, 39, 45]	(6 neighbors, mean seq dist: 18.7)
09 ☰ Small Taming	d=1: [01, 05, 26, 37, 57, 61]	(6 neighbors, mean seq dist: 26.2)
10 ☵ Treading	d=1: [01, 06, 25, 38, 58, 61]	(6 neighbors, mean seq dist: 25.8)
11 ☶ Peace	d=1: [05, 19, 26, 34, 36, 46]	(6 neighbors, mean seq dist: 18.7)
12 ☶ Standstill	d=1: [06, 20, 25, 33, 35, 45]	(6 neighbors, mean seq dist: 17.3)
13 ☱ Fellowship	d=1: [01, 25, 30, 33, 37, 49]	(6 neighbors, mean seq dist: 20.2)
14 ☰ Great Possession	d=1: [01, 26, 30, 34, 38, 50]	(6 neighbors, mean seq dist: 20.2)
15 ☶ Modesty	d=1: [02, 36, 39, 46, 52, 62]	(6 neighbors, mean seq dist: 28.8)
16 ☱ Enthusiasm	d=1: [02, 35, 40, 45, 51, 62]	(6 neighbors, mean seq dist: 27.8)
17 ☱ Following	d=1: [03, 25, 45, 49, 51, 58]	(6 neighbors, mean seq dist: 26.2)
18 ☱ Work on the Decayed	d=1: [04, 26, 46, 50, 52, 57]	(6 neighbors, mean seq dist: 25.8)
19 ☱ Approach	d=1: [07, 11, 24, 41, 54, 60]	(6 neighbors, mean seq dist: 20.5)
20 ☱ Contemplation	d=1: [08, 12, 23, 42, 53, 59]	(6 neighbors, mean seq dist: 19.5)
21 ☱ Biting Through	d=1: [25, 27, 30, 35, 38, 51]	(6 neighbors, mean seq dist: 13.3)
22 ☱ Grace	d=1: [26, 27, 30, 36, 37, 52]	(6 neighbors, mean seq dist: 12.7)
23 ☱ Splitting Apart	d=1: [02, 04, 20, 27, 35, 52]	(6 neighbors, mean seq dist: 14.7)
24 ☱ Return	d=1: [02, 03, 19, 27, 36, 51]	(6 neighbors, mean seq dist: 15.0)
25 ☱ Innocence	d=1: [10, 12, 13, 17, 21, 42]	(6 neighbors, mean seq dist: 11.5)
26 ☱ Great Taming	d=1: [09, 11, 14, 18, 22, 41]	(6 neighbors, mean seq dist: 11.8)
27 ☱ Nourishment	d=1: [21, 22, 23, 24, 41, 42]	(6 neighbors, mean seq dist: 7.8)
28 ☱ Great Preponderance	d=1: [31, 32, 43, 44, 47, 48]	(6 neighbors, mean seq dist: 12.8)
29 ☱ The Abysmal	d=1: [07, 08, 47, 48, 59, 60]	(6 neighbors, mean seq dist: 23.5)
30 ☱ The Clinging	d=1: [13, 14, 21, 22, 55, 56]	(6 neighbors, mean seq dist: 16.8)
31 ☱ Influence	d=1: [28, 33, 39, 45, 49, 62]	(6 neighbors, mean seq dist: 12.7)
32 ☱ Duration	d=1: [28, 34, 40, 46, 50, 62]	(6 neighbors, mean seq dist: 12.7)
33 ☱ Retreat	d=1: [12, 13, 31, 44, 53, 56]	(6 neighbors, mean seq dist: 16.2)
34 ☱ Great Power	d=1: [11, 14, 32, 43, 54, 55]	(6 neighbors, mean seq dist: 15.8)
35 ☱ Progress	d=1: [12, 16, 21, 23, 56, 64]	(6 neighbors, mean seq dist: 19.7)
36 ☱ Darkening of the Light	d=1: [11, 15, 22, 24, 55, 63]	(6 neighbors, mean seq dist: 19.7)
37 ☱ The Family	d=1: [09, 13, 22, 42, 53, 63]	(6 neighbors, mean seq dist: 19.0)
38 ☱ Opposition	d=1: [10, 14, 21, 41, 54, 64]	(6 neighbors, mean seq dist: 19.0)
39 ☱ Obstruction	d=1: [08, 15, 31, 48, 53, 63]	(6 neighbors, mean seq dist: 18.3)
40 ☱ Deliverance	d=1: [07, 16, 32, 47, 54, 64]	(6 neighbors, mean seq dist: 18.3)
41 ☱ Decrease	d=1: [04, 19, 26, 27, 38, 61]	(6 neighbors, mean seq dist: 18.5)
42 ☱ Increase	d=1: [03, 20, 25, 27, 37, 61]	(6 neighbors, mean seq dist: 19.5)
43 ☱ Breakthrough	d=1: [01, 05, 28, 34, 49, 58]	(6 neighbors, mean seq dist: 20.8)
44 ☱ Coming to Meet	d=1: [01, 06, 28, 33, 50, 57]	(6 neighbors, mean seq dist: 21.2)
45 ☱ Gathering Together	d=1: [08, 12, 16, 17, 31, 47]	(6 neighbors, mean seq dist: 23.8)
46 ☱ Pushing Upward	d=1: [07, 11, 15, 18, 32, 48]	(6 neighbors, mean seq dist: 24.8)
47 ☱ Oppression	d=1: [06, 28, 29, 40, 45, 58]	(6 neighbors, mean seq dist: 16.3)
48 ☱ The Well	d=1: [05, 28, 29, 39, 46, 57]	(6 neighbors, mean seq dist: 17.0)
49 ☱ Revolution	d=1: [13, 17, 31, 43, 55, 63]	(6 neighbors, mean seq dist: 18.7)
50 ☱ The Cauldron	d=1: [14, 18, 32, 44, 56, 64]	(6 neighbors, mean seq dist: 18.7)
51 ☱ The Arousing	d=1: [16, 17, 21, 24, 54, 55]	(6 neighbors, mean seq dist: 22.2)
52 ☱ Keeping Still	d=1: [15, 18, 22, 23, 53, 56]	(6 neighbors, mean seq dist: 22.5)

53 ䷗ Development	d=1: [20, 33, 37, 39, 52, 57] (6 neighbors, mean seq dist: 14.7)
54 ䷗ The Marrying Maiden	d=1: [19, 34, 38, 40, 51, 58] (6 neighbors, mean seq dist: 15.3)
55 ䷗ Abundance	d=1: [30, 34, 36, 49, 51, 62] (6 neighbors, mean seq dist: 13.7)
56 ䷗ The Wanderer	d=1: [30, 33, 35, 50, 52, 62] (6 neighbors, mean seq dist: 14.3)
57 ䷗ The Gentle	d=1: [09, 18, 44, 48, 53, 59] (6 neighbors, mean seq dist: 19.2)
58 ䷗ The Joyous	d=1: [10, 17, 43, 47, 54, 60] (6 neighbors, mean seq dist: 20.2)
59 ䷗ Dispersion	d=1: [04, 06, 20, 29, 57, 61] (6 neighbors, mean seq dist: 30.2)
60 ䷗ Limitation	d=1: [03, 05, 19, 29, 58, 61] (6 neighbors, mean seq dist: 31.2)
61 ䷗ Inner Truth	d=1: [09, 10, 41, 42, 59, 60] (6 neighbors, mean seq dist: 24.2)
62 ䷗ Small Preponderance	d=1: [15, 16, 31, 32, 55, 56] (6 neighbors, mean seq dist: 27.8)
63 ䷗ After Completion	d=1: [03, 05, 36, 37, 39, 49] (6 neighbors, mean seq dist: 34.8)
64 ䷗ Before Completion	d=1: [04, 06, 35, 38, 40, 50] (6 neighbors, mean seq dist: 35.2)

--- Neighborhood clustering in King Wen sequence ---

Mean sequence distance between Hamming-1 neighbors: 20.6

Mean sequence distance between any two hexagrams: 21.7

Hamming-1 neighbors are closer than average in the sequence (clustered).

--- Neighborhood clustering null model ---

Shuffling binary hexagrams 10,000 times to build a null distribution
of mean sequence distance between Hamming-1 neighbors.

Null distribution: min=17.9, mean=21.7, max=25.4

King Wen observed mean: 20.6

Percentile: 11.5% (proportion of shuffles with mean <= King Wen's)

King Wen's neighborhood clustering is within the range expected by chance.

Recurrence plot

Recurrence plot

A recurrence plot shows where the difference wave repeats itself. Each cell (i,j)

is marked if the difference at position i equals the difference at position j.

Diagonal lines indicate repeated sequences; vertical/horizontal lines indicate

values that persist. Clusters of marks reveal recurring local patterns.

	1	2	3	4	5	6
12345678901234567890123456789012345678901234567890123						
1	+	#	#	#	#	#
2	+	#	##	###	#	#
3	+	#	#	#	#	#
4	+	#	#	#	#	#
5	+	#	#	#	#	#
6	+	#	#	#	#	#
7	+	#	#	#	#	#
8	+	#	#	#	#	#
9	+	#	#	#	#	#
10	+	#	#	#	#	#
11	+	#	#	#	#	#
12	+	#	#	#	#	#
13	+	#	#	#	#	#
14	+	#	#	#	#	#
15	+	#	#	#	#	#
16	+	#	#	#	#	#
17	+	#	#	#	#	#
18	+	#	#	#	#	#
19	+	#	#	#	#	#
20	+	#	#	#	#	#
21	+	#	#	#	#	#
22	+	#	#	#	#	#
23	+	#	#	#	#	#
24	+	#	#	#	#	#
25	+	#	#	#	#	#
26	+	#	#	#	#	#
27	+	#	#	#	#	#
28	+	#	#	#	#	#
29	+	#	#	#	#	#
30	+	#	#	#	#	#
31	+	#	#	#	#	#
32	+	#	#	#	#	#
33	+	#	#	#	#	#
34	+	#	#	#	#	#

[illegible]

Recurrence rate: 952/3906 (24.4%)

```
--- Recurrence rate null model ---
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Theoretical expected recurrence rate (sum of p_i^2): 25.6%

King Wen recurrence rate: 24.4%

Mean random recurrence rate: 23.3%

King Wen percentile vs random: 72.2%

DNA codon mapping

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DNA codon mapping

Both the I Ching (64 hexagrams) and the genetic code (64 codons) are systems of 64 elements built from binary-like pairs: yin/yang for hexagrams, and the base pairs A-T and G-C for DNA. Each codon is a sequence of 3 bases from the set {A, C, G, U}, giving $4^3 = 64$ possibilities. While this is a numerical coincidence (both are $2^6 = 64$), comparing their structural properties is instructive about combinatorial systems in general.

Caveat: the mapping of bit pairs to DNA bases (00=U, 01=C, 10=A, 11=G) is one of 24 possible assignments. Different mappings produce different codon assignments and different degeneracy statistics. No mapping is privileged by either system. This analysis is illustrative, not evidence of a biological connection.

— — —

Pos	Hex	Binary	Codon	Amino Acid	Name
01	III	111111	GGG	Gly	The Creative
02	III	000000	UUU	Phe	The Receptive
03	III	010001	CUC	Leu	Difficulty at the Beginning
04	III	100010	AUA	Ile	Youthful Folly
05	III	010111	CCG	Pro	Waiting
06	III	111010	GAA	Glu	Conflict
07	III	000010	UUA	Leu	The Army
08	III	010000	CUU	Leu	Holding Together
09	III	110111	GCG	Ala	Small Taming
10	III	111011	GAG	Glu	Treading
11	III	000111	UCG	Ser	Peace
12	III	111000	GAU	Asp	Standstill
13	III	111101	GGC	Gly	Fellowship
14	III	101111	AGG	Arg	Great Possession
15	III	000100	UCU	Ser	Modesty

16		䷮		001000		UAU		Tyr		Enthusiasm
17		䷮		011001		CAC		His		Following
18		䷮		100110		ACA		Thr		Work on the Decayed
19		䷮		000011		UUG		Leu		Approach
20		䷮		110000		GUU		Val		Contemplation
21		䷮		101001		AAC		Asn		Biting Through
22		䷮		100101		ACC		Thr		Grace
23		䷮		100000		AUU		Ile		Splitting Apart
24		䷮		000001		UUC		Phe		Return
25		䷮		111001		GAC		Asp		Innocence
26		䷮		100111		ACG		Thr		Great Taming
27		䷮		100001		AUC		Ile		Nourishment
28		䷮		011110		CGA		Arg		Great Preponderance
29		䷮		010010		CUA		Leu		The Abysmal
30		䷮		101101		AGC		Ser		The Clinging
31		䷮		011100		CGU		Arg		Influence
32		䷮		001110		UGA		Stop		Duration
33		䷮		111100		GGU		Gly		Retreat
34		䷮		001111		UGG		Trp		Great Power
35		䷮		101000		AAU		Asn		Progress
36		䷮		000101		UCC		Ser		Darkening of the Light
37		䷮		110101		GCC		Ala		The Family
38		䷮		101011		AAG		Lys		Opposition
39		䷮		010100		CCU		Pro		Obstruction
40		䷮		001010		UAA		Stop		Deliverance
41		䷮		100011		AUG		Met		Decrease
42		䷮		110001		GUC		Val		Increase
43		䷮		011111		CGG		Arg		Breakthrough
44		䷮		111110		GGA		Gly		Coming to Meet
45		䷮		011000		CAU		His		Gathering Together
46		䷮		000110		UCA		Ser		Pushing Upward
47		䷮		011010		CAA		Gln		Oppression
48		䷮		010110		CCA		Pro		The Well
49		䷮		011101		CGC		Arg		Revolution
50		䷮		101110		AGA		Arg		The Cauldron
51		䷮		001001		UAC		Tyr		The Arousing
52		䷮		100100		ACU		Thr		Keeping Still
53		䷮		110100		GCU		Ala		Development
54		䷮		001011		UAG		Stop		The Marrying Maiden
55		䷮		001101		UGC		Cys		Abundance
56		䷮		101100		AGU		Ser		The Wanderer
57		䷮		110110		GCA		Ala		The Gentle
58		䷮		011011		CAG		Gln		The Joyous
59		䷮		110010		GUA		Val		Dispersion
60		䷮		010011		CUG		Leu		Limitation
61		䷮		110011		GUG		Val		Inner Truth
62		䷮		001100		UGU		Cys		Small Preponderance
63		䷮		010101		CCC		Pro		After Completion
64		䷮		101010		AAA		Lys		Before Completion

Unique amino acids in King Wen order: 21
Total possible amino acids + stop: 21

--- Degeneracy comparison ---
In DNA, many single-base changes preserve the amino acid (the code is 'degenerate').
Do single-line hexagram changes preserve the mapped amino acid?
Single-line changes that preserve amino acid: 100/384 (26.0%)

Shannon entropy analysis

Shannon entropy analysis
Entropy measures disorder: high entropy means the difference values are spread evenly across all possibilities (random-looking), low entropy means certain values dominate (structured). We compare the King Wen wave's entropy against thousands of random permutations of the same 64 hexagrams. If King Wen's entropy is unusually low, the sequence is more structured than random chance would produce.

King Wen difference wave entropy: 2.0759 bits
Maximum entropy (all 7 values): 2.8074 bits
Maximum entropy (5 observed values): 2.3219 bits

Mean entropy of random permutations: 2.1926 bits
Min random entropy observed: 1.6006 bits
Max random entropy observed: 2.4818 bits
King Wen percentile: 13.5% (lower = more structured)
Effect size (Cohen's d): -1.12 (negative = more structured than random)

--- Entropy conditioned on pair constraint ---
The unconstrained comparison above may be misleading: the pair structure itself constrains the entropy. How does King Wen compare against random orderings that also satisfy the pair constraint?
Mean pair-constrained entropy: 2.2342 bits
King Wen percentile (pair-constrained): 5.6%
(Similar to unconstrained percentile of 13.5%.)

--- Distribution comparison ---

Value	King Wen	Expected (random avg)
0	0	0.0
1	2	6.0
2	20	15.0
3	13	20.0
4	19	14.9
5	0	6.1
6	9	1.0

Path analysis (graph theory)

Path analysis (graph theory)
Imagine the 64 hexagrams as cities on a map, where the 'distance' between any two is the number of lines that differ. The King Wen sequence is a route that visits all 64 cities. Is it a short, efficient route (nearby hexagrams placed next to each other), or a long, wandering one? We compare its total distance against random routes and a greedy 'always go to the nearest unvisited city' algorithm. This reveals whether the King Wen ordering was optimized for smooth transitions or had other priorities.

King Wen total path length: 211 (sum of all line changes)
Greedy nearest-neighbor: 75
Mean random path length: 192.0
Min random observed: 158
Max random observed: 230
King Wen percentile: 97.6% (lower = shorter path)
Effect size (Cohen's d): +2.05

--- Theoretical bounds ---
Minimum possible (63 transitions of 1): 63
Maximum possible (63 transitions of 6): 378
King Wen uses 55.8% of maximum path length

--- Pair-constrained comparison ---
The above compares against fully random orderings. A fairer comparison is against random orderings that also preserve the pair structure (each pair of hexagrams stays adjacent). This is the right null model for asking whether King Wen's path length is unusual GIVEN its pair constraint.
Mean pair-constrained path length: 214.1
Min pair-constrained observed: 189
Max pair-constrained observed: 237
King Wen percentile (pair-constrained): 28.6%
Effect size (Cohen's d, pair-constrained): -0.48

Constraint satisfaction analysis

Constraint satisfaction analysis
The King Wen sequence satisfies several constraints simultaneously:

1. All 32 consecutive pairs are either reverse or inverse (never unrelated)
2. No consecutive hexagrams differ by exactly 5 lines
3. No consecutive hexagrams are identical (0-line transitions)

How rare is it for a random ordering to satisfy all these at once? We test random permutations against each constraint individually and combined.

Results from 10,000 random permutations:

```
All pairs reverse/inverse:      0 (0.000%)
No 5-line transitions:         21 (0.21%)
Both constraints together:      0 (0.0000%)
No random permutation satisfied both constraints.
Statistical note: 0/10,000 gives a 95% upper bound of <0.0300%
(i.e., the true rate is less than ~1 in 3,333, but we cannot
distinguish 'impossible' from 'extremely rare' with this sample size).
```

--- Conditional probability: P(no-5 | pair constraint) ---

The pair structure constrains transitions within pairs (always even or 6), so 5-line transitions can only occur at the 31 between-pair boundaries.

How often do pair-constrained orderings also avoid 5-line transitions?

```
Pair-constrained trials: 100,000
Also satisfy no-5:       4,231 (4.23%)
Approximately 1 in 23 pair-constrained orderings avoid 5-line transitions.
The no-5 property is uncommon but not extraordinary among pair-constrained orderings.
```

--- Sensitivity analysis: reversed bit convention ---

What if bit 0 = top line instead of bottom? All binary values reverse, changing pair types and the difference wave. Key properties tested:

```
Pair structure preserved:  Yes
No-5 property preserved:   Yes
Difference wave identical: Yes
Original entropy: 2.0759, reversed: 2.0759
Original path length: 211, reversed: 211
All key properties are invariant under bit reversal.
```

Bootstrap confidence intervals

Bootstrap confidence intervals

The Monte Carlo results (e.g., '0.18% of random orderings avoid 5-line transitions') are estimates based on sampling. Bootstrap resampling quantifies how precise those estimates are by repeatedly resampling from the results and computing confidence intervals. Narrower intervals = more reliable estimates.

```
Base trials: 10,000
Bootstrap resamples: 1000
```

```
No-5-line-transition rate: 0.170%
95% confidence interval: [0.100%, 0.260%]
Interval width: 0.160 percentage points
```

```
Approximately 1 in 588 random orderings
95% CI: 1 in 385 to 1 in 1000
```

Note: These CIs measure the precision of the Monte Carlo estimate, not fundamental uncertainty about the true proportion. They reflect how much the estimate would vary if you re-ran the simulation with 10,000 trials. Increasing --trials narrows these CIs because the estimate becomes more precise.

Monte Carlo analysis

Monte Carlo analysis (10,000 random permutations)

The King Wen sequence has a striking property: no two consecutive hexagrams differ by exactly 5 lines. With 6 lines per hexagram and 7 possible difference values (0-6), is avoiding 5 remarkable or just a coincidence? To find out, we randomly shuffle the 64 hexagrams thousands of times and check how often a random ordering also avoids 5-line transitions. The rarer it is, the more likely the King Wen sequence was intentionally designed with this constraint.

```
Permutations with no 5-line transitions: 21/10,000 (0.21%)
Approximately 1 in 476 random orderings share this property.
```

Odds ratio against random: 475:1